



Parul University Faculty of Engineering and Technology Department of Applied Science & Humanities

Academic Year 2024-25

Subject: Quant and Reasoning (203105371)

Branch: CSE

Unit: 8 – Clocks and Calendars Calendar Reasoning

Calendar Reasoning Tricks, Notes, Questions

Calendars Calendar Reasoning

Tricks, Notes, Questions Days and Week

One week includes following 7 Days :

Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday

The days repeat itself after completion of week.

Odd Day

The remaining days after calculated Completed weeks are odd days. It can be find by divide the number of days from the 7, remainder is Odd Days.

Days of the week start from Monday (1 Jan 0001 was Monday) , Therefore, 1 odd day means Monday.

Odd Day	1	2	3	4	5	6	7/0
Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun

Months	Days	Odd Days
Jan, March, May, July, August, October, December	31	3
February Normal Year	28	0

February Leap Year	29	1
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April, June, September, November	30	2
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Tricks – Odd days in a Week -‘0 ‘

Add or Subtract 7 from any day, you will find same day.
 Like 1 July 2020 is Wed, 8 July 2020 ?
 $1+7 = 8$, Therefore, 8 July 2020 is Wednesday

Odd days in Months Normal

Year

- Normal year, which is not divisible by 4 is a Normal Year, However in Century year like 100, 200, 1900, 2000, it should not be divisible by 400.
- That means 1900 is divisible by 4 and we may think it a leap year, but it is century year (last two digits 00), we have to divide it by 400.
Hence, it is Normal Year.
- Normal Year = 365 day = 52 weeks + 1 day (1 Odd Day)
February – 28 days

Leap Year

Leap Year = 366 days = 52 weeks + 2 day (2 odd days)
 February – 29 days

Example : Which of the following is a leap year
 (a) 2021, (b) 2022 (c) 2023 (d) 2024

Answer : (d) 2024 is divisible by 4, therefore it is a Leap Year

Tricks Calendar Reasoning: First and Last day of a year

In a Normal Year

1 st day of the year = Last days of the year
 1 Jan and 31 Dec will be same day
 Example: 1 Jan 2022 is Saturday, and 31 Dec 2022 is also Saturday

What will be 1 Jan 2023 ?

Normal Year Sat +1 = Sunday

In Leap Year

31 Dec is 1+ 1st Jan Day

Example 1 Jan 2024 is Monday, 31 Dec 2024 = Monday+1 =Tuesday

After 400 Years Calendar repeat itself

1 Jan 2001 – Monday
1 Jan 2401 will Monday
1 Jan 1601 was Monday ... etc

Odd days in Century and Last Day

Century	Odd Day	Last Day 31Dec	First Day 1Jan
100, 500, 900, 1300, 1700, 2100..	5 OD	Friday	Monday
200,600, 1000, 1400, 1800, 2200 ..	$5+5=10 = 3 \text{ OD}$	Wednesday	Saturday
300,700,1100,1500,1900,2300..	$5+5+5 = 15 = 1 \text{ OD}$	Monday	Thursday
400, 800,1200, 1600,2000,2400.. Century Leap Year	$5+5+5+6 = 21 = 0 \text{ OD}$	Sunday	Tuesday

Example Questions : Type wise

Type 1: Day and Month same Year Different

12 March 2018 is Sunday. What will be 12 March 2022 ?

Solution :

2018 NY -1
2019 NY -1
2020 LY- 2
2021 NY-1

OD -5 , Sunday +5 = Friday

Trick : 2022- 2018 = 4 Year + 1 LY = 5

23 March 1835 is Sunday. What will be 23 March 1882 ?

Solution

$1882-1835 = 47 \text{ years} + 12 \text{ LY} = 59/7 = 3 \text{ OD}$
Sunday + 3 = Wednesday

NOTE : To calculate LY : LY after 1835 is 1836
LY after 1884
 $1884 - 1836 = 48/4 = 12 \text{ LY}$

Type -2 : Date and Year Same Month Different

04 March 2011 is Sunday. What will be 04 August 2011 ?

Solution

Odd day of Mar -3, Apr-2, May-3, Jun-2, July-3 = $13/7 = 6 \text{ OD}$
Sunday +6 = Saturday

Note : August OD will not be counted.

Type 3: Month and year Same Date Different

04 March 2022 is Friday. What will be 30 March 2022 ?

Number of days = $30-4 = 26/7 = 5 \text{ OD}$
Friday +5 = Wednesday

Note : Friday (5) +5 = $10/7 = 3 \text{ (Wed)}$

Type 4: Date, Month, Year all different

Example 1: 11 July 2020 is Saturday, what will be 22 October 2028 ?

Years : $2028 - 2020 = \text{Year- } 8 + \text{LY- } 2 = 10$, OD -3
Month : July - 3, Aug-3, Sep-2 =8, OD -1 (Oct will not be counted)
Date : $22-11 = 11$, OD = 4
Total OD = $3+1+4 = 8$, OD =1

Therefore, Sat +1 = Sunday

Example 2: 15 March 2021 is Monday, What will be 12 July 2023.

Year : $2023-2021 = 2 + 0$ (LY) = 2 OD

Months : March ($31-15 = 16$) -2 OD, Apr-2, May-3, June-2, July 12 – 5 OD

Months OD : $2 + 2 + 3 + 2 + 5 = 14 = 0$ OD

Total OD : Year + Months = $2 + 0 = 2$ OD

Therefore 12 July 2023 will be Monday + 2= Wednesday

Example 3 : 14 November 2020 is Saturday, What will be the day of 26 January 2026.

Month/Year	Day	Odd Days
Nov 2020	$30-14=16$	2
Dec 2020	31	3
2021 Full Year NY	365	1
2022 NY	365	1
2023 NY	365	1
2024 LY	366	2
2025 NY	365	1
Jan 2026	26	5
		$16 = 2$ OD

Therefore, 26 January 2026 = Saturday + 2 = Monday

Type- 5: Find the Day of a back date from the given date and day

When the day of a date is given and day of a date previous to it is asked. We will subtract the odd days from the given day.

Example: 26 January 2020 is on Sunday, What was the day on 26 January 1950.

Years : $2020 - 1950 = \text{year } 70 + \text{LY } 17 = 87 - 3 \text{ OD}$

(Including – 2000 CLY,)

Note: For LY 2020 (Before Feb) – $1952 (\text{NLY}) = 68/4 = 17$

26 January 1950 was Sunday (0/7) – 3 = **Thursday**

Note : We will subtract the Odd days from the day

Example: 10 Feb 2020 is on Monday, What was the day on 25 December 1918.

Dec 2018, $31 - 25 = 6 \text{ OD}$

Year 2019 NY = 01 OD

Jan 2020 31 days = 3 OD

Feb 2020 , 10 days = 3 OD

Total OD = $6 + 1 + 3 + 3 = 13 = 6 \text{ OD}$

25 December 1918 was (Monday – 6) = **Tuesday**

Note : Monday $1 + 7 - 6 = 2$ nd day of week

Find the day of a given date

Example 1 : What was the day on 15 August 1947

Completed Year upto 15 Aug 1947 = $1900 + 46$

Century (1900 years) Odd days = **1 OD**

$46 (\text{Years}) + 11 (\text{LY}) = 57 = 1 \text{ OD}$

Completed Months of 1947 (NY): Jan-3, Feb-0, Mar-3, Apr-2, May-3, Jun-2, Jul-3,

Total OD = 16 = **2 OD**

Day of August-15 = **1 OD**

Total OD = Century –1 + 46 Years –1 + Months 2 + Days – 1 = **5 OD 15**

August 1947 was Friday (5 th Day of week)

(Note : Odd day for 300,700,1100,1500,1900,2300.. Century is 1 and Number of LY – 1948 – 1904 = 44/4 =11)

Example 2: Birth date of a student is 25 March 2004, Find the day of that day.

Completed Year : 2000 + 3

Century multiple of 400, code is 0, LY -0

Odd Day = 0 (C Code) + 3 (years) + 0 (LY) =3 OD

Completed Months of 2004 (LY): Jan-3, Feb-1 , = 4 OD

Days 25 = 4 OD

Total OD = 3+4+4 = 11 = 4 OD

25 March 2004 was the Thursday (4 th day of week)

Extra Questions

Q 1: Ravi went to see a Movie, 9 day before. He see the Movie only on Friday. Today is what Day of week.

Today = Friday + 9 = Friday + 2 OD = **Sunday**

Q 2: 29 Feb is Monday. what day was on 11 February.

29, 22, 15 and 8 Feb Monday

8+3 = 11, Monday+3 = **Thursday**

Q 3 : If this year is a Leap Year and 1 January is Monday. After How Many year 1 Jan will be on Monday.

This Year LY – 1 Jan – Mon

After 1 Year NY – 1 Jan Wed (2 OD of previous Year)

After 2 Year NY – 1 Jan Thu (1 OD of previous Year)

After 3 year NY – 1 Jan Fri (1 OD of previous Year)

After 4 Year LY – 1 Jan Sat (1 OD of previous Year)

After 5 Year NY – 1 Jan Monday (2 OD of previous Year)

After 5 Year – (The total of Odd days is 7 Means Monday)

Alternate Method

Add the Odd day of years till it become 7 or multiple of 7.

$$2 \text{ LY} + 1 \text{ NY} + 1 \text{ NY} + 1 \text{ NY} + 2 \text{ LY} = 7 \text{ (After 5 years)}$$

Q 4: Ram celebrated his 4th birthday on 29 Feb 2020. When he was born.

4 Birthday 29 Feb 2020

3 Birthday 29 Feb 2016

2 Birthday 29 Feb 2012

Ram Birth Date 29 Feb 2008

Solved example

1. It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2010?

- A. Sunday
- B. Saturday
- C. Friday
- D. Wednesday

Answer: Option C

2. What was the day of the week on 28th May, 2006?

- A. Thursday
- B. Friday
- C. Saturday
- D. Sunday

3. What was the day of the week on 17th June, 1998?

- A. Monday
- B. Tuesday
- C. Wednesday
- D. Thursday

Answer: Option c

4. What will be the day of the week 15th August, 2010?

- A. Sunday
- B. Monday
- C. Tuesday
- D. Friday

Answer: Option A

Explanation:

5. Today is Monday. After 61 days, it will be:

- A. Wednesday
- B. Saturday
- C. Tuesday
- D. Thursday

Answer: Option B

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6. If 6th March, 2005 is Monday, what was the day of the week on 6th March, 2004?

- A. Sunday
- B. Saturday
- C. Tuesday
- D. Wednesday

Answer: Option A

7. On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?

- A. Tuesday
- B. Monday
- C. Sunday
- D. Wednesday

Answer: Option C

8. Which of the following is not a leap year?

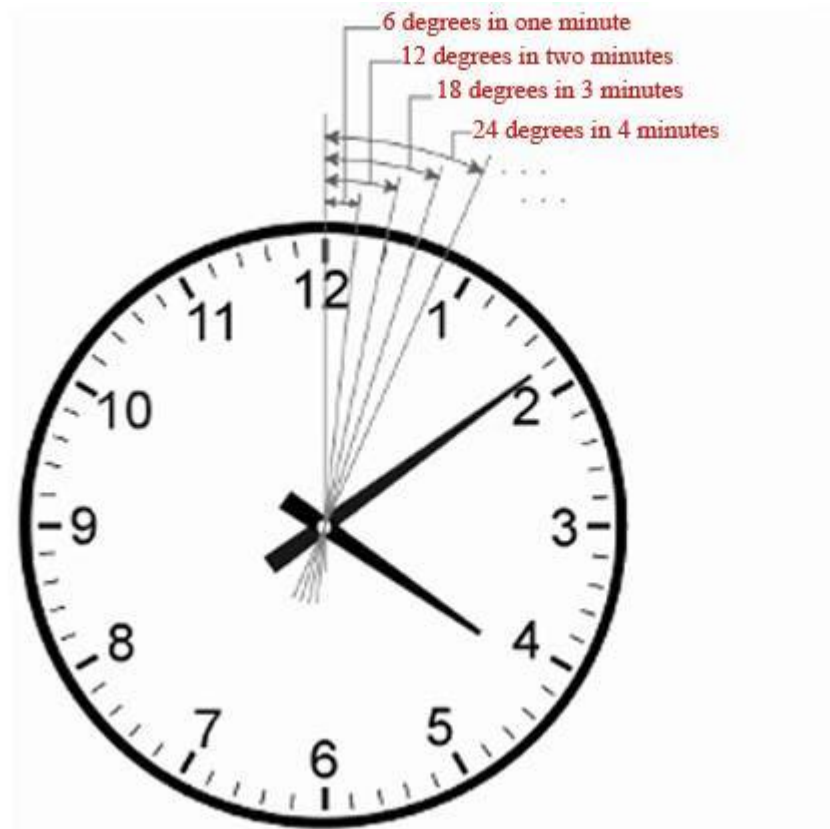
- A. 700
- B. 800
- C. 1200
- D. 2000

Answer: Option A

Clock Reasoning Formula, Tricks, Questions

Clock Formula, Tricks, Example,

Clock Reasoning Formula



Speed of Minute Hand (MH)

60 Minutes – 360 Degree
 1 Minute – $360/60 = 6$ Degree

Speed of Hour hand (HH)

12 hrs = 720 Minutes – 360 Degree
 1 hr = $360/12 = 30$ Degree
 1 Minute = $360/720 = \frac{1}{2}$ Degree

Difference between Hour and Minute Hand in 1 Minute = $6 - \frac{1}{2} = 5 \frac{1}{2} = 11/2$ Degree

Minute and Hr Both Hands	Angle in Degree	Happened in 12 hrs
Coincident (ठीक ऊपर)	0	11
Right Angle	90	22

Opposite	180	11
Straight Line	0 or 180	22

Find the Angle between Hour and Minute hand at a given Time

Example 1: Find angle between the Hrs and Minute hands at 7 hrs and 20 Minutes

The HH will be ahead of 7.
 Upto 7 HH travel $7 \times 30 = 210$ degree
 The angle travel by HH ahead of 7.
 $\frac{1}{2}$ degree in every minute, and $20 \times \frac{1}{2} = 10$ degree
 $210 + 10 = 220$ Degree

MH is exactly at 20 Minutes
 Means travel $20 \times 6 = 120$ degree
 The Angle is $220 - 120 = 100$ Degree

Tricks: We can use the relative (Difference) speed of HH and MH i.e. $6 - \frac{1}{2} = \frac{11}{2}$ Degree

Formula : $\text{HH} \times 30 (-) \text{MH} \times \frac{11}{2}$

$7 \times 30 - \frac{11}{2} \times 20 = 210 - 110 = 100$ degree

Example 2 : Find the Angle at 2 hrs and 40 Minutes

$2 \times 30 - 40 \times \frac{11}{2} = 60 - 220 = 160$ degree

Minutes Hand travel in 1 hr = 60 Minutes
 Hour hand travel in 1 hr = 5 Minute
 Therefore, Minutes Hand travel $60 - 5 = 55$ Minutes more than HH.

Example: Between 3 and 4 when the clock Hands will be coincident

MH travel in 55 Minutes more in 60 Minutes
 MH travel in 1 Minutes more in $\frac{60}{55}$ Minutes
 MH travel in 15 Minutes = $15 \times \frac{60}{55} = \frac{180}{11} = 16 \frac{4}{11}$ Minutes
 3 hrs and $16 \frac{4}{11}$ Minutes

Trick Formula: $\text{T hrs and Minutes} \times \frac{12}{11}$

3 hrs and $15 \times \frac{12}{11} = \frac{180}{11} = 16 \frac{4}{11}$ Minutes

T is First Time from the Given Time.

For Right Angle (90 degree), 2 time in every hr $T : 60/11 \times (T \pm 3)$

For any other angle we will check the Completed Minutes upto First Time

Example : Between 9 and 10, when both Clock Needles, were in straight line, but not opposite.

Both Clock Needles are Coincident after 9.

Time is 9 Hrs and $45 \times 12/11 = 9$ hrs and $49 \frac{1}{11}$ Minutes.

Example: Between 9 and 10, when both Clock Hands are opposite.

MH completed 15 Minutes

Time is 9 hrs and $15 \times 12/11 = 9$ hrs and $16 \frac{4}{11}$ Minutes

Example: Between 6 and 7, when the Clock hands will made Right Angle Triangle.

There will be two Right Angle (90 Degree)

Case 1 : Minute hand just ahead of 15 Minute

6 hrs and $15 \times 12/11 = 16 \frac{4}{11}$ Minutes

Case 2 : MH just after 45 Minutes

6 hrs and $45 \times 12/11 = 49 \frac{1}{11}$ Minutes

Mirror Image of a Clock

Trick Subtract the given time from 11:60 or 23:60

Example: Time in a watch is 10:20, what will be the time in Mirror Image.

$$11:60 - 10:20 = 1:40$$

Ex. : Mirror Image of a watch at 12:10.

दर्पण छवि 12:10 पर।

$$23:60 - 12:10 = 11:50$$

Ex: Mirror Image of a Clock shows time 06:30. What is the actual time ?

$$11:60 - 06:30 = 05 : 30$$

Water Image of Clock

Formula : Subtract the given time from 17:90

Ex. A wall clock having dots in place of number. That Clock is placed over water. The time is 8:50 in water image. What is the actual time.

Actual time is $17:90 - 08:50 = 09:40$

Example 2: If the water image of the clock shows 4 hours 25 minutes, then what will be the actual time?

Solution: For every water image based on a clock, If minutes given are less than 30, then subtract given timing from 18:30 hours. Same as 17:90

And if minutes given are more than 30, then subtract given timing from 18:90 hrs.

Here, minutes are less than 30, hence, $18:30 - 4:25 = 14:05$ i.e. 2 hours 5 minutes.

Thus, the actual time shows 2 hours 5 minutes in a clock.

Types of question that can be asked

- 1) The angle between the hands of the clock
- 2) Time at a particular angle
- 3) Position of hands of the clock
- 4) Faulty clocks
- 5) Time gained or lost by the clock
- 6) Mirror / Water Image-based time

Now, let's discuss each of them in detail:

Type 1: Angle between the hands of the clock

Example 1: What will be the angle made by the hands of the clock at 8:15?

FORMULA: $|(60H - 11M) / 2|$ or : $(HH \times 30) - (MH \times 11/2)$

Solution: Angle at 8:15

$(60 \times 8 - 11 \times 15) / 2$

$(480 - 165) / 2 = 315 / 2 = 157.5$ degree

TRICK: Hour hand $\rightarrow$ 1 hour = 30 degree

8 hour $\rightarrow$ 240 degree

1 minute = $1/2$ degree

15 minute = 7.5 degree

Minute hand $\rightarrow$ 1 minute = 6 degree

15 minute = 90 degree

Now, Angle made by hour hand – Angle made by minute hand
Required Angle = $240 + 7.5 - 90 = 247.5 - 90 = 157.5$ degree

Type 2: Time at a particular angle

Example 1: When will the 90-degree angle be formed between 7:00 and 8:00?

FORMULA: Angle between the two hands = $|(60H - 11M) / 2|$

ie $90 = 11/2M - 30H = 11/2M - 30 \times 7$

ie $90 + 210 = 11/2M$

$300 = 11/2M$, $M = 600/11 = 54$, and $6/11$. past 7

Now, to find the 2nd right angle between 7 and 8,

$90 = 30H - 11/2M = 30 \times 7 - 11/2M$

$11/2M = 210 - 90 = 120$, hence $M = 240/11 = 21$ and $9/11$ min. past 7

OR

Trick Formula: (T hrs) and (Minutes X 12/11)

T = starting time ie. 3 and Minutes = two hands together after two is when minute hand will be at 2 ie 10 minutes

(T hrs) and (Minutes X 12/11) = (7 hrs) and (20 X 12/11) = 2 and $21 \frac{9}{11}$ minutes

Type 3: Position of hands of the clock

Example 1: How many times a day do the hands of the clock make a 180-degree angle?

Needles meet each other

	Degree	In a day (24 hours)	12 hours
	0 degree / 360 degree	22 times	11 times
	180 degree	22 times	11 times
	90 degrees	44 times	22 times

Solution: Thus, 22 times a day hands of the clock make a 180-degree angle.

1. If the hands of the clock are inclined at 15 minutes past 5, then what was the angle of the hands?

A. $59 \frac{1}{2}^\circ$ B. 65° C. $66 \frac{1}{2}^\circ$ **D. $67 \frac{1}{2}^\circ$**

15 minutes past 5 mean at 5:15, 5 hour and 15 minutes

$$(HH \times 30) - (MH \times 11/2) = (5 \times 30) - (15 \times 11/2) = 150 - 82.5 = 67.5$$

2. The reflex angle between the hands of the clock at 10:25 AM is

- A. 180° B. 185° C. 150° D. $197 \frac{1}{2}^\circ$

$$(HH \times 30) - (MH \times 11/2) = (10 \times 30) - (25 \times 11/2) = (300 - 137.5) = 162.5$$

The reflex angle is the large angle formed by the hand of clock since full rotation is 360 degree then reflex angle can be formed by subtracting the acute angle

$$\text{reflex angle} = 360 - \text{acute angle} = 360 - 162.5 = 197 \frac{1}{2}^\circ$$

3. Suppose a clock started at noon. When the minute hand is 10 past 5 than the hour hand has turned through

- A. 140° B. 155° C. 160° D. 170°

10 past 5 mean at 5:10, 5 hour and 10 minutes

$$\text{Angle between two hand} = (5 \times 30) - (10 \times 11/2) = (150) - (55) = 95$$

At 5:10 minute hand is at 2 so angle between 12 and 2 is 60

So hour hand has turned through $95 + 60 = 155$

4. What will be the angle between the minute hand and the hour hand of the clock when the time is 4:50 PM?

A. 155° B. 10° C. 25° D. 45°

Angle between two hand = $(4 \times 30) - (50 \times 11/2) = (120) - (275) = 155$

5. Find the angle between the hour hand and the minute hand of a clock when the time is 3:25 AM.

A. 46° B. $47 \frac{1}{2}^\circ$ C. 50° D. 55°

Angle between two hand = $(3 \times 30) - (25 \times 11/2) = (90) - (137.5) = 47.5 = 47 \frac{1}{2}^\circ$

6. Find the time between 2 and 3 o'clock when the hands of a clock will be together.

A. 2 and $10 \frac{9}{11}$ minutes B. 2 and $10 \frac{10}{11}$ minutes C. 2 and 8 minutes D. 2 and $10 \frac{6}{11}$ minutes

Trick Formula: (T hrs) and (Minutes X 12/11)

T = starting time ie. 3 and Minutes = two hands together after two is when minute hand will be at 2 ie 10 minutes

(T hrs) and (Minutes X 12/11) = (2 hrs) and (10 X 12/11) = 2 and $10 \frac{10}{11}$ minutes

Find the Answers Here

1. D. $67\frac{1}{2}^\circ$
2. D. $197\frac{1}{2}^\circ$
3. B. 155°
4. A. 155°

5. B. $47\frac{1}{2}^\circ$

6 B. 10 $\frac{10}{11}$ minute

Part 2: Clock Test Practice Questions

In the below questions, you need to find the exact time when the clock is slow or fast or the exact time between.

1. A watch which gains uniformly is 5 minutes slow at 8 o'clock in the morning on Monday and it is 5 min. 48 sec fast at 8 pm the following Monday. When was it correct?

- A. 20 min past 6 on Tuesday
- B. 20 min past 6 on Wednesday
- C. 20 min past 7 on Wednesday
- D. 20 min past 7 on Thursday

2. At what time between 5 and 6 o'clock are the hands of a clock 3 minutes apart?

A. 20 min B. 24 min C. 26 min D. 30 min

3. How many times in one day will the hands of the clock coincide?

A. 22 0 B. 21 C. 20 D. 24

4. How many times in the hands of the clock at a right angle in one day?

A. 24 B. 22 C. 48 D. 44

5. At what time between 530 and 6 will the hands of a clock be at the right angle?

A. 40 min B. $43 \frac{7}{11}$ in C. $43 \frac{5}{11}$ min D. 45 min

6. At what time between 9 and 10 o'clock will the hands of a clock be together?

A. 45 min B. 40 min C. $48 \frac{2}{11}$ min D. $49 \frac{1}{11}$ in

Find the Answers Here

1. C. 20 min past 7 on Wednesday

2. 24 min

3. 22

4. 44

5. $43 \frac{7}{11}$ min

6. $49 \frac{1}{11}$ min

1.

An accurate clock shows 8 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?

- A. 144°
- B. 150°
- C. 168°
- D. 180°

Answer: Option D

Explanation:

Angle traced by the hour hand in 6 hours $= 30 \times 6 = 180$

2.

The reflex angle between the hands of a clock at 10.25 is:

- A. 180°
- B. $192\frac{1}{2}^\circ$
- C. 195°
- D. $197\frac{1}{2}^\circ$

Answer: Option D

Explanation:

$$(\text{HH} \times 30) - (\text{MH} \times 11/2) = (10 \times 30) - (25 \times 11/2) = (300 - 137.5) = 162.5$$

$$\text{reflex angle} = 360 - \text{acute angle} = 360 - 162.5 = 197\frac{1}{2}^\circ$$

OR

$$\text{Angle traced by hour hand in } \frac{125}{12} \text{ hrs} = \left(\frac{360}{12} \times \frac{125}{12} \right)^\circ = 312\frac{1}{2}^\circ$$

$$\text{Angle traced by minute hand in 25 min} = \left(\frac{360}{12} \times 25 \right)^\circ = 150^\circ$$

$$\text{Reflex angle} = 360^\circ - \left(312\frac{1}{2}^\circ - 150^\circ \right) = 197\frac{1}{2}^\circ$$

3. A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:

- A. 145°
- B. 150°
- C. 155°
- D. 160°

Answer: Option C

Explanation:

Angle between two hands at 4:50 = $(4 \times 30) - (50 \times 11/2) = (120) - (275) = 155$

Angle traced by hour hand in 12 hrs = 360° .

Angle traced by hour hand in 5 hrs 10 min. i.e. $\frac{31}{6}$ hrs = $\left(\frac{360}{12} \times \frac{31}{6}\right)^\circ = 155^\circ$